Exploratory Analysis of Global Earthquake-Tsunami Events (2001–2022)

Introduction:

Earthquakes are natural seismic events caused by movements of tectonic plates in the Earth's crust. Some earthquakes generate tsunamis, which can cause significant damage in coastal regions. Understanding earthquake patterns and tsunami risk is important for disaster preparedness.

This project performs **Exploratory Data Analysis (EDA)** on a global earthquake dataset from **2001 to 2022**. The aim is to analyze earthquake characteristics such as magnitude, depth, and geographic location, and study their relationship with tsunami generation using Python visualization techniques.

Objectives:

The main objectives of this project are:

1. To analyze earthquake occurrences and tsunami events over time.
2. To study the distribution of earthquake magnitude and depth.
3. To visualize earthquake locations using latitude and longitude.
4. To compare seismic characteristics between tsunami and non-tsunami earthquakes.
5. To derive insights about factors associated with tsunami generation.

Tools and Libraries Used:

The analysis was conducted using Python with the following libraries:

- **Pandas** – data loading and manipulation
- **NumPy** – numerical operations
- **Matplotlib** – data visualization
- **Seaborn** – statistical visualization

The project was implemented using **Jupyter Notebook**.

Data Analysis:

Several visualizations were created to analyze the dataset:

- **Time-based analysis** showed how earthquake and tsunami events vary across different years.
- **Magnitude and depth analysis** helped understand the distribution of earthquake strength and underground depth.
- **Geographic scatter plots** displayed earthquake locations based on latitude and longitude.
- **Statistical comparisons** using histograms and box plots helped compare tsunami and non-tsunami earthquakes.
- **Correlation heatmaps** were used to study relationships between seismic variables.

Harshal Patil

Key Insights:

- Earthquake occurrences vary across different years.
- Tsunami-generating earthquakes are relatively rare compared to total earthquakes.
- Higher magnitude earthquakes are more likely to generate tsunamis.
- Earthquake locations show clusters in tectonically active regions.
- Magnitude and depth appear to influence tsunami occurrence.

Conclusion:

The exploratory analysis of earthquake data from **2001–2022** reveals patterns in seismic activity and tsunami generation. The study shows that earthquakes with higher magnitudes have a greater probability of generating tsunamis. Visualization techniques help identify trends and relationships between different seismic variables. These insights contribute to better understanding of earthquake behavior and tsunami risk.

Harshal Patil